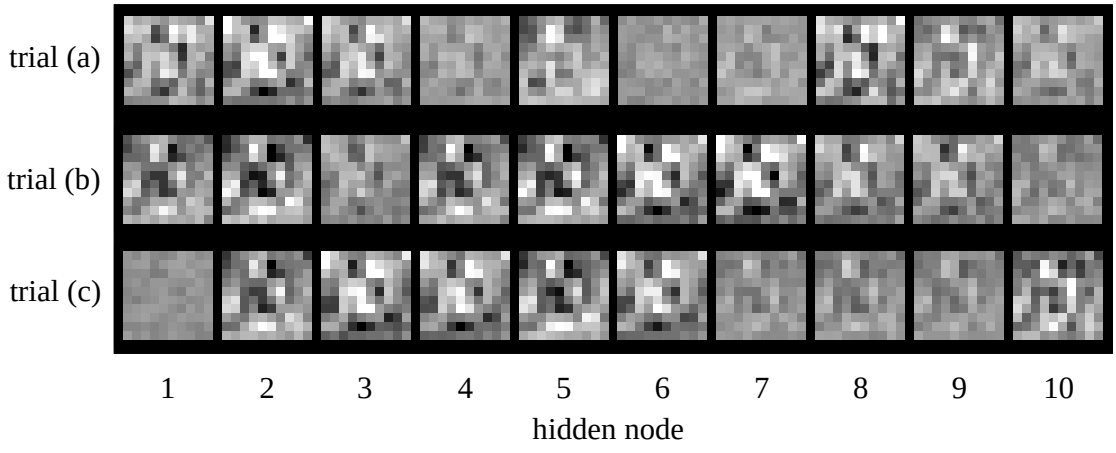


FIGURE 26-10
Example of neural network weights. In this figure, the hidden layer weights for the three solutions are displayed as images. All three of these solutions appear random to the human eye.

In Fig. 26-10, the hidden layer weights of the three solutions are displayed as images. This means the first action taken by the neural network is to correlate (multiply and sum) these images with the input signal. They look like random noise! These weights values can be *shown* to work, but *why* they work is something of a mystery. Here is something else to ponder. The human brain is composed of about 100 *trillion* neurons, each with an average of 10,000 interconnections. If we can't understand the simple neural network in this example, how can we study something that is at least 100,000,000,000,000 times more complex? This is 21st century research.

Figure 26-11a shows a histogram of the neural network's output for the 260 letters in the training set. Remember, the weights were selected to make the output near *one* for vowel images, and near *zero* otherwise. Separation has been perfectly achieved, with no overlap between the two distributions. Also notice that the vowel distribution is narrower than the nonvowel distribution. This is because we declared the target error to be five times more important than the nontarget error (see line 2220).

In comparison, Fig. 26-11b shows the histogram for images 261 through 1300 in the database. While the target and nontarget distributions are reasonably distinct, they are not completely separated. Why does the neural network perform better on the first 260 letters than the last 1040? Figure (a) is cheating! It's easy to take a test if you have already seen the answers. In other words, the neural network is recognizing specific images in the training set, not the general patterns identifying vowels from nonvowels.

Figure 26-12 shows the performance of the three solutions, displayed as ROC curves. Trial (b) provides a significantly better network than the